$$W_n = u_{n+1}v_n - u_n v_{n+1}$$

Insert the shared recurrence
The Q_n terms cancel

$$W_n = -(C_n/A_n)W_{n-1}$$

$$W_n = \frac{(-1)^{n+1}p(n+1)}{2(2n+1)^2(2n+2)^2}$$

with $p(t) = 20t^2 - 8t + 1$